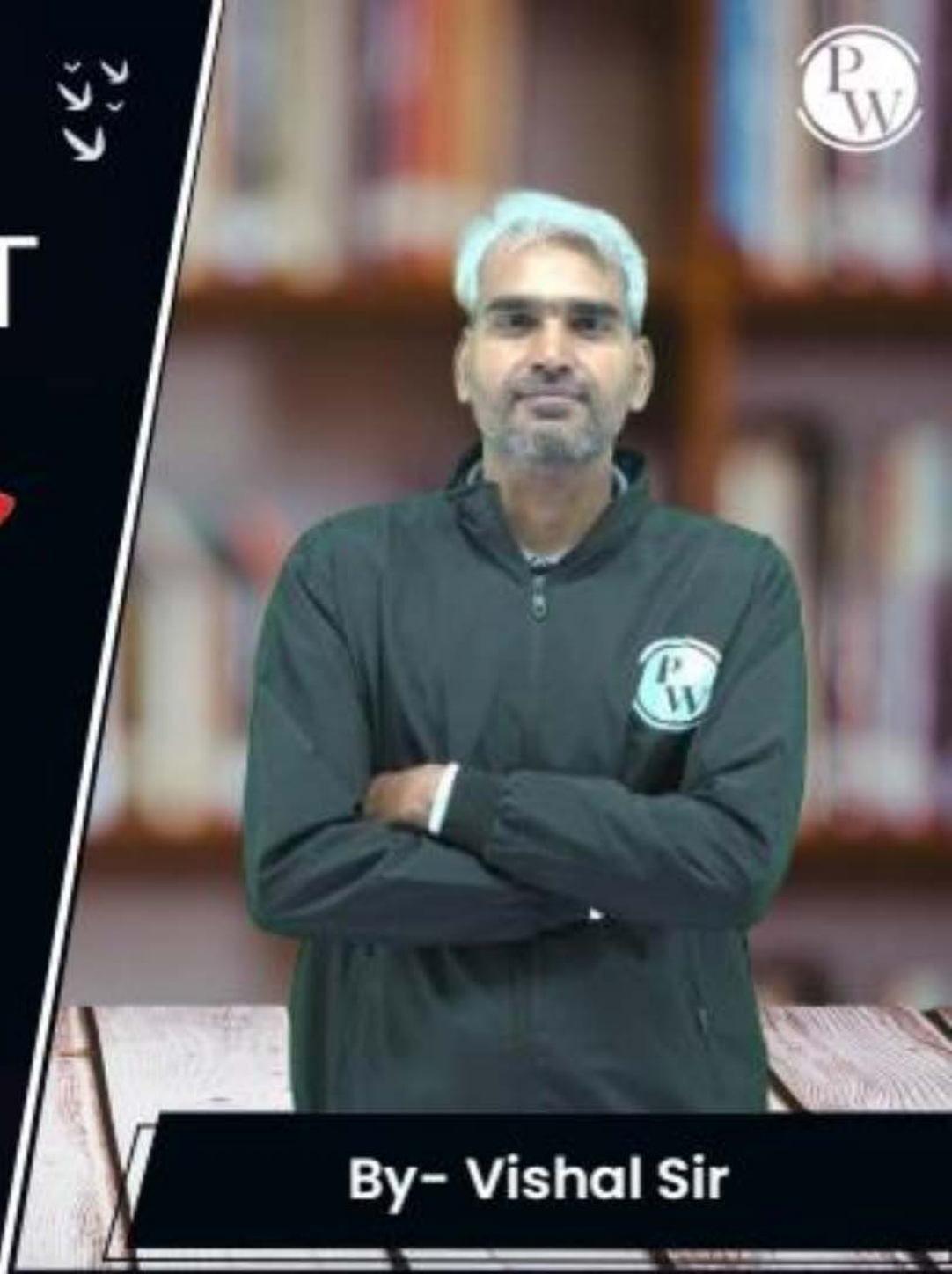
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture



Topic SLR(1) Parser

Topic Conflict conditions w.r.t. SLR(1) Parser

Slide

Topics to be Covered



Topic Need of LR(1) items

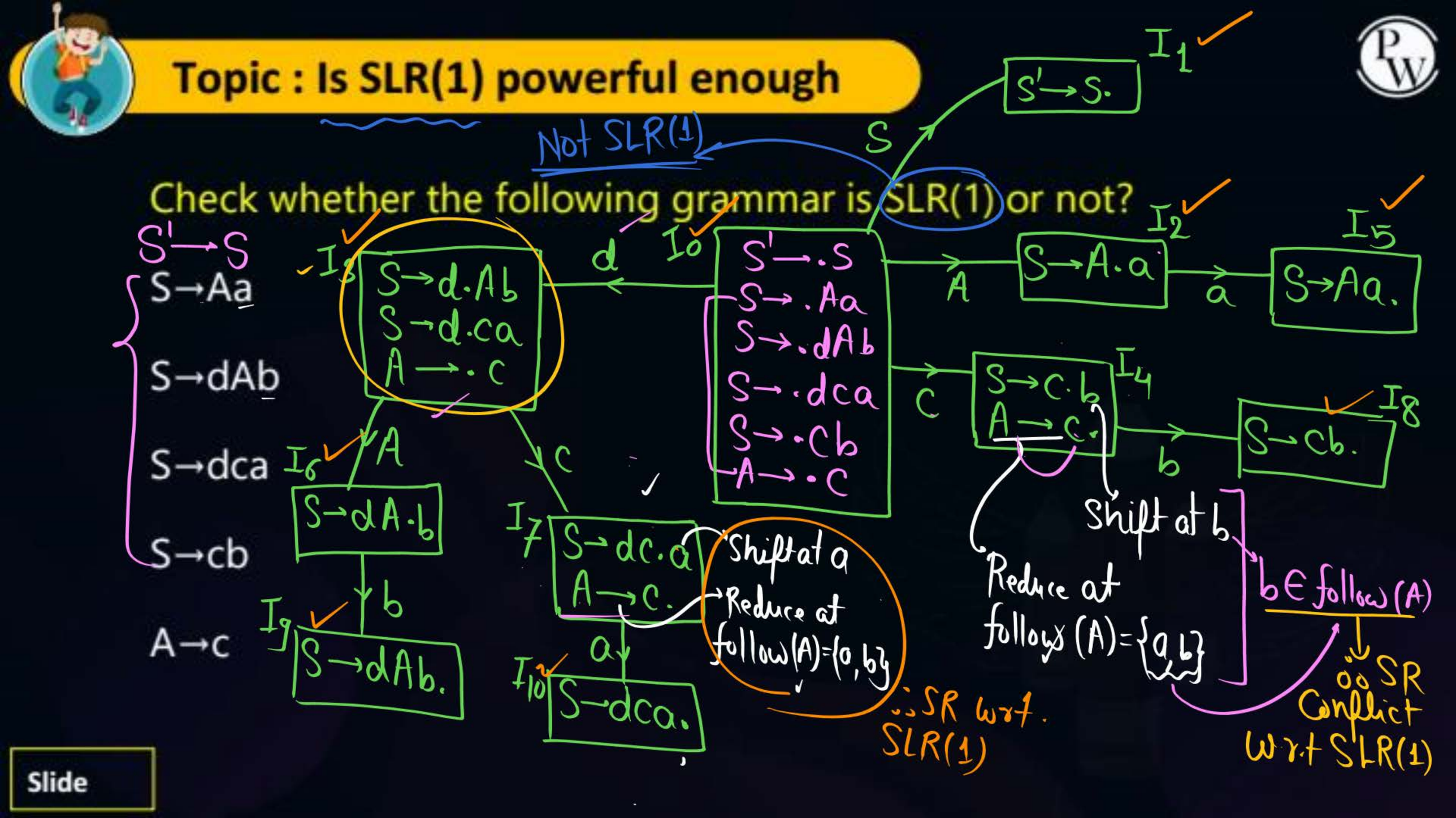
Topic Construction of DFA of LR(1) items

Topic : Is SLR(1) powerful enough

Not SLR(1)

Check whether the following grammar is SLR(1) or not?

$S' \to S$
$S \to Aa$
$S \to dAb$
$S \to dca$
$S \to cb$
$A \to c$



Slide



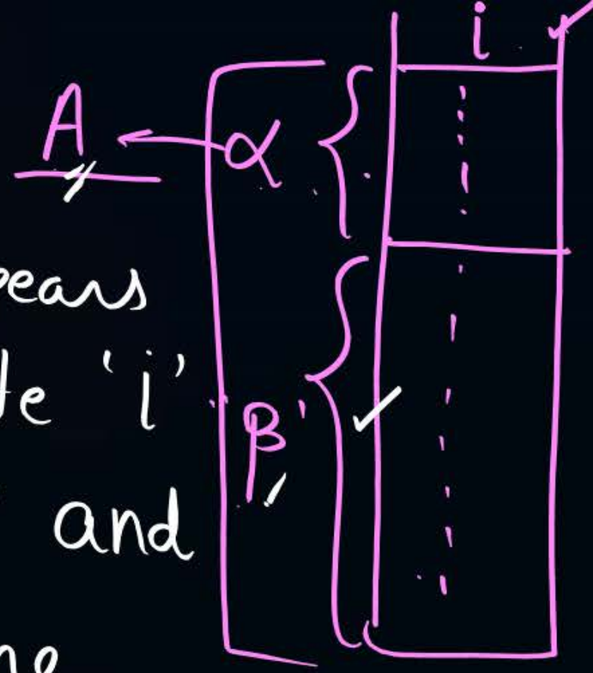
Topic : Need of LR(1) items

* In LR(0) no look-ahead is used to place reduce entry in LR(0) parsing table.
* In SLR(1) look-ahead may be determined by Computing the follow set of the non-terminal w.r.t. which reduction will be performed

Slide



Topic : Need of LR(1) items



'a' ∈ follow(A)

In SLR(1) method, Let state 'i' appears on the top of the stack, If state 'i' calls for the reduction by 'A → α' and terminal a ∈ follow(A), then in some situation when state 'i' appears on top of the stack the Viable Prefix "βα" in the stack may be such that if we reduce α into A, then "βA" can not be followed by 'a', in such situation reduction of α by A will be invalid if next input symbol is 'a'.

Available in Stack

Sequence of symbols that we have already Parsed

Slide



Topic : Need of LR(1) items

- It may be possible to incorporate some more information into the state that will allow us to rule out some of this invalid reductions by $A \rightarrow \alpha$

Slide



Topic : Need of LR(1) items

The extra information can be incorporated into the states by redefining the items to include a terminal of length '1'.

∴ the general form of the item becomes

$[A \rightarrow \alpha . \beta, t]$

't' is a terminal

$A \rightarrow \alpha . \beta$: same as LR(0) item

t : look-ahead of length '1'

Such items are called LR(1) items

Slide

LR(1) item = LR(0) item + Look-ahead of length '1'

Look ahead length

Note:- ① The look-ahead has no effect if LR(1) item is of the form

$[A \longrightarrow \alpha . \beta, a]$ where $\beta \neq \epsilon$ { i.e. item is not w.r.t. reduce action }

a any terminal

② But when $\beta = \epsilon$, i.e. when the item is of the form

$[A \longrightarrow \alpha ., a]$, then it calls for reduction by $A \rightarrow \alpha$ only if the next input symbol is terminal 'a'.



Topic : Construction of DFA of LR(1) items

Step 1: Construct the initial state

Step 2:- Start a new state

Step 3:- Compute the newly started state {i.e. obtain closure}

Step 4:- Repeat step ② & step ③ until no new state can be constructed.

Slide



Topic : Initial state Construction

(a) If $S' \longrightarrow S$ is the augmented production then add $[S' \longrightarrow .S, \$]$ into the initial state.

(b) If there exists any item of the form $[A \longrightarrow \alpha.B\gamma, t]$ in the state, and $B \longrightarrow \beta$ is a production in the grammar, then add $[B \longrightarrow .\beta, u]$ into the state where $u \in first(\gamma t)$

(c) Repeat step (b) until no other item can be added into the state

Slide



α, β can be anything

Topic : Starting a new state

Let $[A \rightarrow \alpha . X\beta, t]$ is a item in state 'i' s.t. $X=(V+T)$, then we locate every item of the type in which Symbol X appear immediately on right side of DOT (.) mark, and then we start a new state 'j' by moving the DOT (.) mark after X in all those items,

in state 'i'

Note: Look-ahead of the item does not change when we start a new state



Topic : Closure of newly started state

- For every item of the type $[A \rightarrow \alpha . B\gamma, t]$ in the newly started state,

  (B: non-terminal)

  add $[B \rightarrow .\beta, u]$ in the state

  where $u \in first(\gamma t)$

  * Repeat for every newly added item if non-terminal follows DOT(.) mark

Note: look-ahead may change bcoz of closure

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Topic : LR(1) item

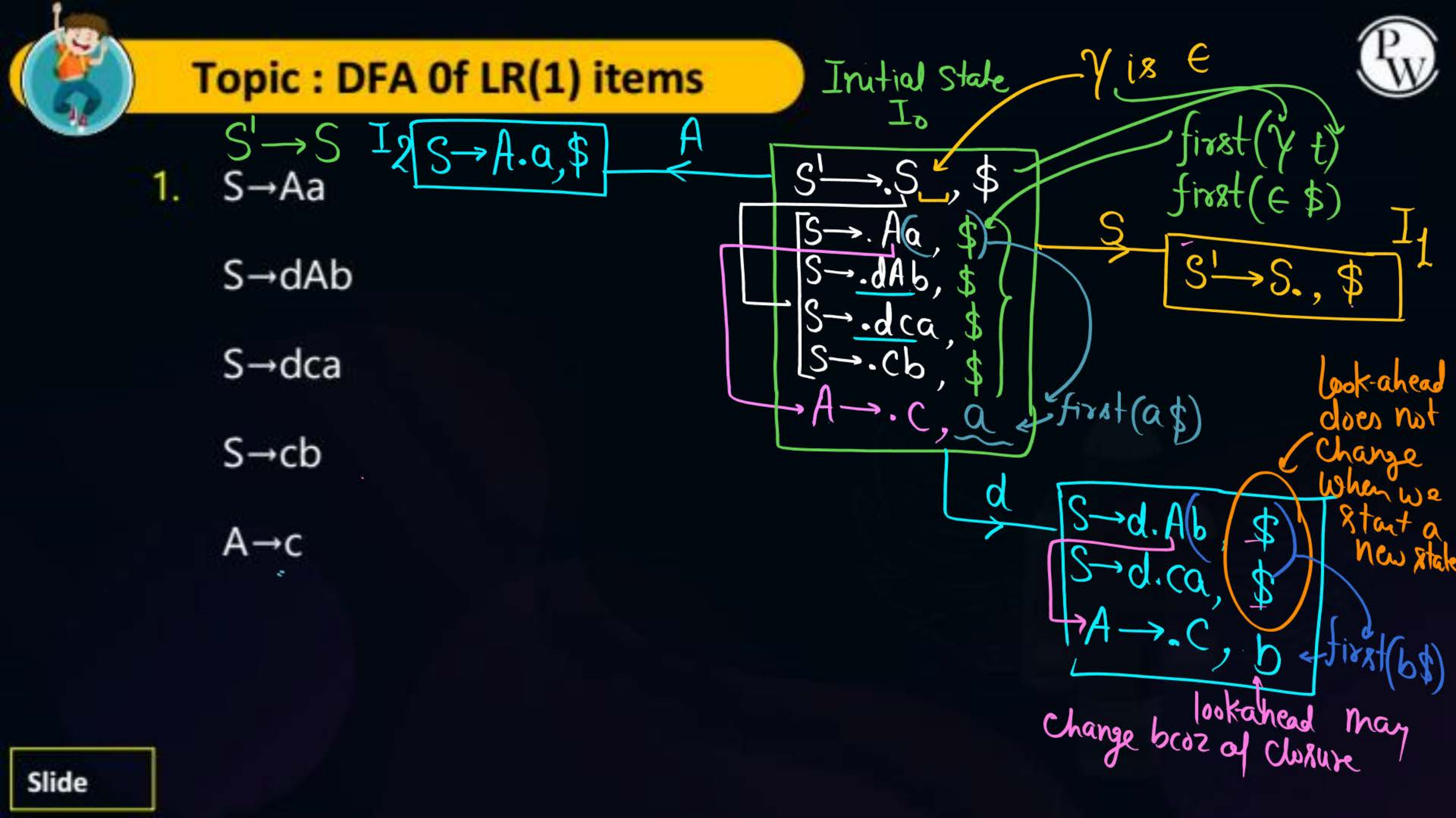
LR(1) item = LR(0) item + look ahead of length '1'

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Topic : DFA Of LR(1) items

$S' \rightarrow S$

1. $S \rightarrow Aa$
   $S \rightarrow dAb$
   $S \rightarrow dca$
   $S \rightarrow cb$
   $A \rightarrow c$





Topic : DFA 0f LR(1) items

2. $S \rightarrow C\ C$

$C \rightarrow e\ C$

$C \rightarrow d$

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Topic : DFA 0f LR(1) items

3. $S \rightarrow A\ A$

$S \rightarrow bAc$

$S \rightarrow dc$

$S \rightarrow bda$

$A \rightarrow d$

Slide



Topic : DFA 0f LR(1) items

4. $S \rightarrow A\ a$

$S \rightarrow b\ A\ c$

$S \rightarrow Bc$

$S \rightarrow b\ B\ a$

$A \rightarrow d$

$B \rightarrow d$

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2 mins Summary

Topic Need of LR(1) items

Topic Construction of DFA of LR(1) items

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THANK - YOU

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